

Experiment:19

Design a C program to implement process synchronization using mutex locks.

Code:-

```
#include <stdio.h>

#include <pthread.h>

int counter = 0;

pthread_mutex_t mutex;

void *threadFunction(void *arg) { for (int i = 0; i < 1000000; i++)
{ pthread_mutex_lock(&mutex); counter++; pthread_mutex_unlock(&mutex); }

return NULL;}

int main() { pthread_t thread1, thread2;

pthread_mutex_init(&mutex, NULL);

pthread_create(&thread1, NULL, threadFunction, NULL);
pthread_create(&thread2, NULL, threadFunction, NULL);

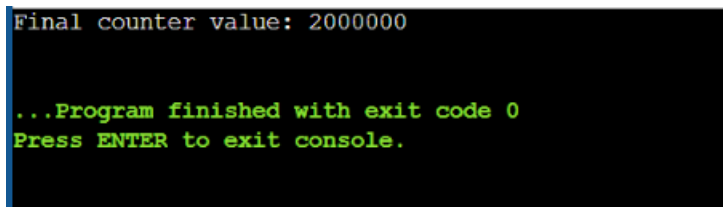
pthread_join(thread1, NULL);
pthread_join(thread2, NULL);
printf("Final counter value: %d\n", counter);

pthread_mutex_destroy(&mutex);

return 0;

}
```

Output:-



```
Final counter value: 2000000

...Program finished with exit code 0
Press ENTER to exit console.
```

